

# ARC: Approach for identifying proxies of demographic correction

Generate Reference Cohort

Select variables relevant to outcome of interest

Identify variables that explain population differences

Develop and evaluate models

## GLI-Global 2022

## ARC<sub>PFT</sub>

### Dataset



GLI Dataset



UK BioBank



NHANES

### Target

Reference Pulmonary Function

FEV1

FVC

FEV1 / FVC

### Feature Selection

Age

Sex

Height

Individual-Level Features

### Algorithm

Generalized Linear Model  
w/ Inverse Probability Weights

XGBoost

### Evaluation

Comparison to  
Race-Specific Equations

GLI-2012

GLI-Global

Group-Wise Performance

White

Black

Mexican American

Comparison to Race-Neutral  
and Race-Specific Equations

GLI-Global

GLI-Global 2022

Group-Wise Performance

*In-Distribution*

White

Black

Asian

Other

*Out of  
Distribution*

White

Black

Asian

Other

Hispanic

## Identifying Individual Features

### Domain-Wide Association Analysis

Anthropometrics



SDoH

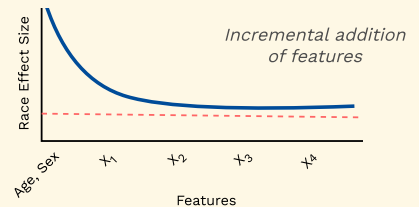


Exposures



### Explain Race-Related Differences

$PFT \sim \text{Age, Sex, Race, *Features}$



### Quantify Predictive Value of Each Feature

Understand what race is proxying in  
race-reference equations